Experiment 3: ASCII Code to HEX Code Conversion Using 8085

AIM:

To write an assembly language program to convert an ASCII code into its corresponding binary (HEX) value using the 8085 microprocessor.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

ALGORITHM:

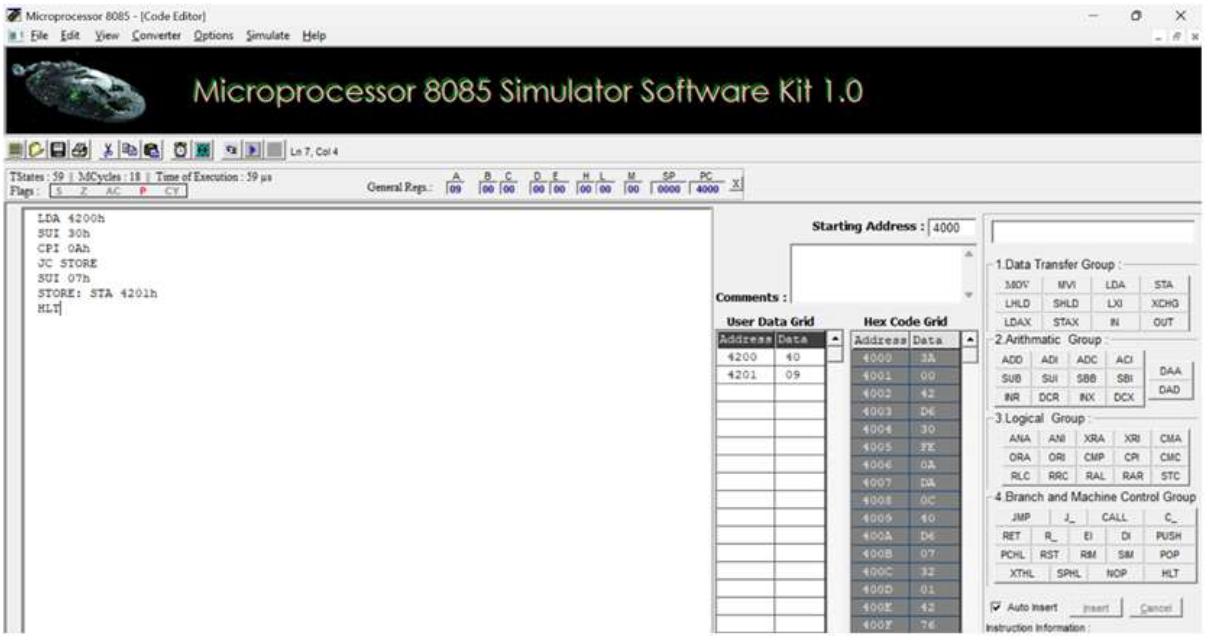
1. Load the ASCII data from memory location 4200H into the accumulator.
2. Subtract 30H from the accumulator.
3. Compare the accumulator with 0AH.
4. If the carry flag is set, skip the next subtraction.
5. Otherwise, subtract 07H from the accumulator.
6. Store the HEX value in memory location 4202H.
7. Stop the program.

PROGRAM:

```
LDA 4200H
SUI 30H
CPI 0AH
JC SKIP
SUI 07H

SKIP: STA 4202H
HLT
```

OUTPUT:



RESULT:

Thus, an assembly language program to convert an ASCII code into its corresponding HEX value was written and executed using the 8085 microprocessor.